



**LANDMARK METROPOLITAN
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THE PRIDE OF AFRICA

FUNCTIONS AND MODULES

SE309 – Python Programming with Project

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FUNCTIONS



Functions are **reusable** blocks of code that perform specific tasks. They help organize code, reduce repetition, and make programs easier to understand.

A function only runs when it is called.

Functions are defined using the **def** keyword, followed by a function name, parentheses **()**, and a colon **:**.

The body of the function contains code it will execute.

```
def greet():  
    print("Hello, welcome!")
```

FUNCTIONS



- To execute a function, simply call it by its name followed by parentheses.

`greet()`

- Functions can return values using the return keyword, which allows them to pass results back to the part of the program that called them

```
def add(a, b):  
    return a + b
```

```
result = add(3, 5)
```

PARAMETERS AND ARGUMENTS



Parameters are variables listed inside the function's parentheses in the definition, used as placeholders for values that the function will use.

```
def greet(name):  
    print(f"Hello, {name}!")
```

Arguments are the actual values passed to the function when it's called.

```
greet("Alice")
```

Types of Parameters

- **Positional Parameters:** These parameters depend on the order of arguments passed
- **Keyword Parameters:** These allow passing arguments with the name of the parameter, not just position.

```
def introduce(name, age):  
    print(f"{name} is {age} years old.")
```

```
introduce(age=30, name="Alice")
```

PARAMETERS AND ARGUMENTS



- **Default Parameters:** Functions can have default values for parameters, which are used if no argument is provided.

```
def greet(name="Guest"):
    print(f"Hello, {name}!")
```

```
greet()
```

MODULES AND PACKAGES



Modules are files containing Python code (functions, variables, etc.) that can be imported into other programs.

Modules help organize code into separate files for easy maintenance and reusability.

Using Built-in Modules

Python provides many built-in modules like math and random for common tasks. For example:

```
import math
```

```
print(math.sqrt(16)) # Outputs: 4.0
```

A package is a collection of modules grouped together, usually in a directory with an `_init_.py` file. Packages organize related modules, making it easier to structure larger programs

IMPORTING MODULES



Entire modules or specific functions can be imported for example:

- **Import random #importing the entire module**
- **from random import randint #importing randint function from random**
print(randint(1, 10))

NOTE: Besides importing built-in modules, we can also create our own modules. This is very useful if you have some functions that you want to reuse in other programming projects in future.

Simply save the file with a .py extension and put it in the same folder as the python file you are going to import it from.